



Cambridge (CIE) IGCSE Computer Science



Your notes

Types of Software & Interrupts

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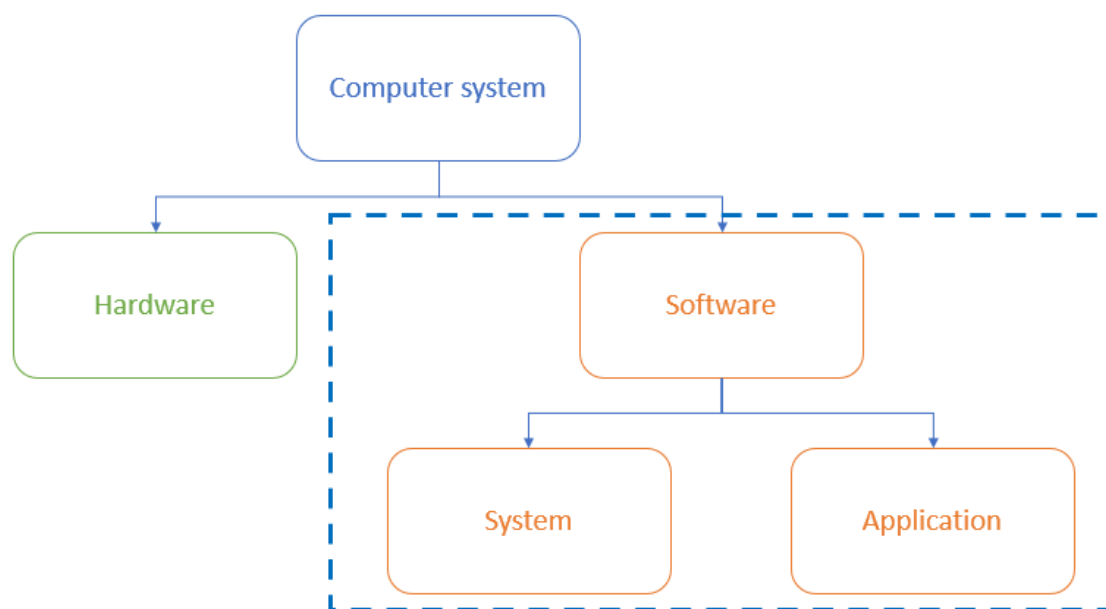


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System Software & Application Software

System Software & Application Software

- Software can be broken down into two categories, **system** and **application** software



What is system software?

- System software is software **essential for the operation of the computer system**
- It gives users **a platform to run applications** and **carry out tasks**
- Examples of system software include:
 - [The operating system](#)
 - Utility software**

Utility software

- Utility software is software designed to help **maintain, enhance** and **troubleshoot/repair** a computer system
- Designed to **perform a limited number of tasks**

- **Interacts with the computers hardware**, for example, secondary storage devices
- Some utility software comes **installed with the operating system**
- Examples of utility software and their function are:
 - **Defragmentation** (maintain)
 - **Compression** (enhance)
 - **Encryption** (enhance)
 - **Task manager** (troubleshoot/repair)

What is application software?

- Application software (abbreviated 'apps') is software **chosen by a user** to help them **carry out a specific task**
- **Installed on top of system software** and is user chosen to best suit **individual requirements**
- Common categories of application software include:
 - **Productivity** - get things done efficiently (word processors, spreadsheets & presentation)
 - **Communication** - stay connected (email, browser, messaging)
 - **Entertainment** - Watch movies, play games or listen to music



Examiner Tips and Tricks

Do not use **brand names** when talking about software, it is ok to talk about software categories or software types but brand names do not get awarded marks

e.g. word processors instead in Microsoft Word



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The Purpose & Functionality of Operating Systems

What is an operating system?

- An operating system (OS) is software that **manages computer hardware** and **provides a platform for running applications**
- It provides **an interface between the user and the hardware** in a computer system
- It **hides the complexities of the hardware** from the user, for example:
 - A user does not need to know 'where' on secondary storage data is kept, just that it is saved for when they want it again
- An operating systems **main functions** can be divided in to **eight** key areas

File Management

What is file management?

- File management is a process carried out by the operating system **creating, organising, manipulating and accessing files and folders on a computer system**
- The OS **manages where data is stored** in both **primary and secondary storage**
- File management gives the user the ability to:
 - **Create files/folders**
 - **Name files/folders**
 - **Rename files/folders**
 - **Copy files/folders**
 - **Move files/folders**
 - **Delete files/folders**
- The OS allows users to control who can access, modify and delete files/folders (**permissions**)
- The OS provides a **search facility** to find specific files based on various criteria

Handling Interrupts

What is interrupt handling?

- Interrupt events **require the immediate attention** of the **central processing unit**

- In order to maintain the smooth running of the system, interrupts need to be **handled** and processed in **a timely manner**
- For example, if a user clicks cancel on a file conversion process, a signal is sent from the mouse, **interrupts the processor**, and the operating system will trigger the cancellation routine



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User Interface

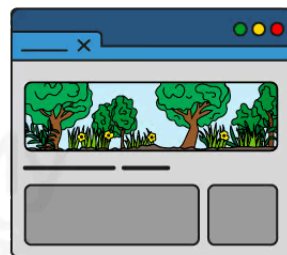
What is a user interface?

- A user interface is **how the user interacts with the operating system**
- Examples of user interfaces include:
 - **Command Line Interface (CLI)**
 - **Graphical User Interface (GUI)**
 - **Menu**
 - **Natural language (NLI)**



CLI

VS.



GUI

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What is a command line interface?

- A Command Line Interface (CLI) requires users to interact with the operating system using **text based commands**
- CLIs are more commonly used by **advanced users**
- Examples of CLIs are MSDOS (Microsoft Disk Operating System) and Raspbian (for Raspberry Pi)

What is a graphical user interface?



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- A Graphical User Interface (**GUI**) requires users to interact with the operating system using **visual elements** such as windows, icons, menus & pointers (**WIMP**)
- GUIs are **optimised for mouse and touch gesture input**
- Examples of GUIs are Windows, Android and MAC OS

What is a menu interface?

- A menu interface is **successive menus** presented to a user with a **single option at each stage**
- Often performed with **buttons** or a **keypad**
- Examples include
 - **Chip and pin machines**
 - **Vending machines**
 - **Entertainment streaming services**

What is a natural language interface?

- A natural language interface (**NLI**) uses the spoken word to respond to spoken or textual inputs from a user
- Examples include
 - **Virtual assistants** – Amazon Alexa, Google Assistant, Siri
 - **Search engines**
 - **Smart home devices**

Advantages and disadvantages of user interfaces

Interface	Advantages	Disadvantages
Command line (CLI)	▪ Uses less system resources ▪ Useful for automation of tasks ▪ Commands are often faster to type than navigating menus	▪ Requires users to remember commands ▪ Typing errors are common ▪ Less intuitive than GUI
Graphical (GUI)	▪ Intuitive and user-friendly ▪ Requires no previous knowledge to use	▪ Uses more system resources ▪ Can be slower to find and execute commands



	Information is visual, making it easier to understand	Can be frustrating when doing repetitive tasks
Menu	- Simplicity - Efficiency	- Limited flexibility - Accessibility issues
Natural language (NLI)	- Can be used by people with disabilities - Intuitive	- Not always reliable - Privacy concerns

Peripheral Management & Device Drivers

What is peripheral management?

- Peripheral management is a process carried out by the operating system **managing the way peripherals (hardware) interact with software**
- The OS **allocates system resources** to peripherals to ensure **efficient operation**
- Peripheral management makes **plug-and-play (PnP)** functionality possible, **automatically detecting and configuring new peripherals** without the need for manually installing device drivers or power cycling the system

What is a device driver?

- A device driver is **a piece of software used to control a piece of hardware**
- Peripherals **require device drivers** in order to be used by the operating system
- The OS has **generic device drivers built in** which makes **basic compatibility** possible and enables **plug-and-play (PnP)**
- In order for hardware to be used to its **maximum capacity**, often **a separate device driver must be downloaded** from the manufacturer
- Device drivers are **OS specific** and are **regularly updated**

Memory Management & Multitasking

What is memory management?

- Memory management is a process carried out by the operating system **allocating main memory (RAM) between different programs** that are **open at the same time**



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- The OS is responsible for **copying programs and data from secondary to primary storage** as it is needed
- **Programs and data require different amounts of RAM** to operate efficiently and the OS manages this process
- **RAM is allocated** based on **priority and fairness**, for example, system applications (essential) may have a higher priority than user applications
- The OS **dynamically manages the memory**, adjusting allocation as needed to maintain optimal system performance
- Memory management makes **multitasking** possible

What is multitasking?

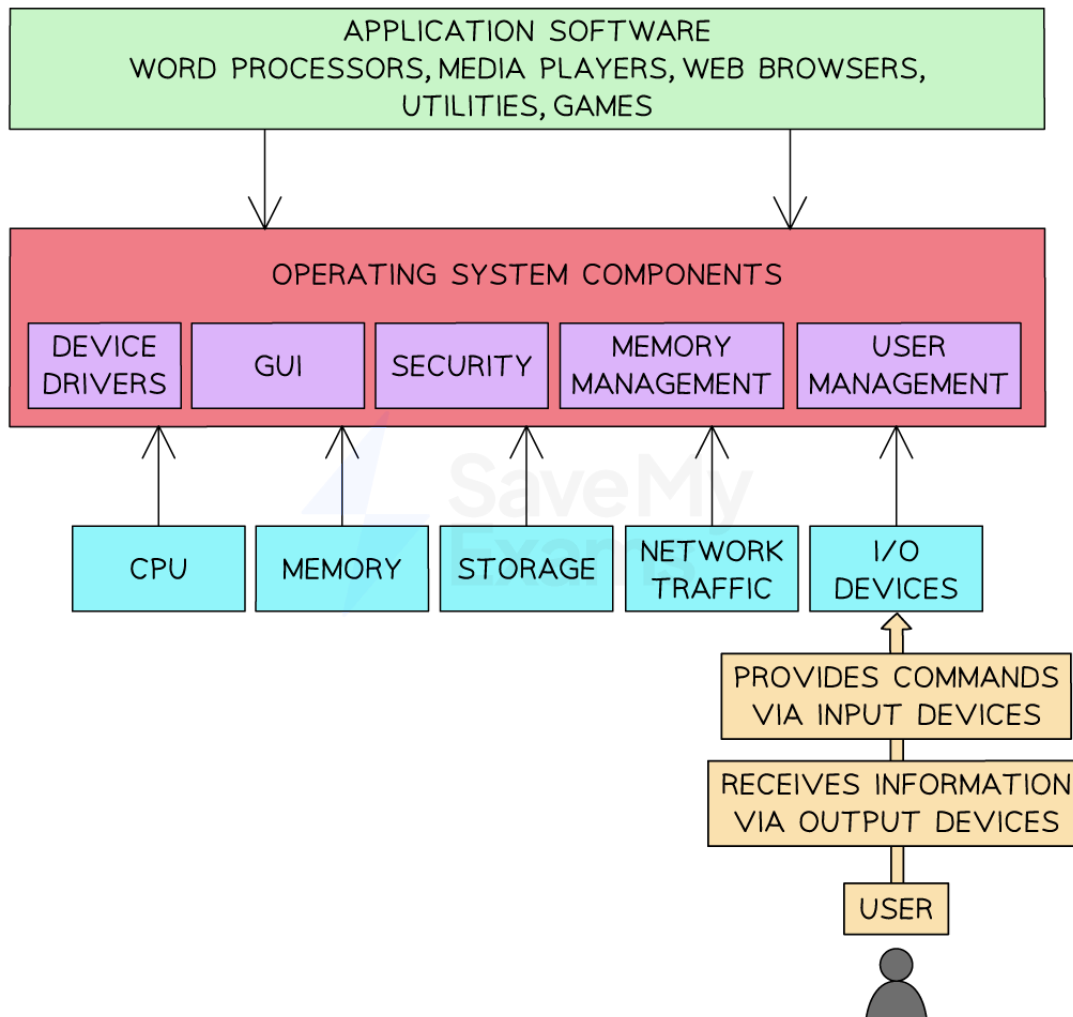
- Multitasking is a process made possible by the OS **simultaneously managing system resources** (memory, CPU etc) to give a user the **perception of being able to use multiple programs at the same time**
- The OS **splits tasks and allocates system resources** based on a priority
- The CPU can only **execute one instruction at a time**, it can execute billions of them in one second.
- This makes it **appear that multiple programs are running at the same time**

Providing a Platform for Running Applications

- Operating systems provide a **platform** on which **application software can run**, this is mainly by allowing software **access to system resources**
- For example, if a computer game has intensive graphics and online play, the operating system will grant it access to the GPU and the network card



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Organisation of application layer, operating system components, and input/output

Providing System Security

What is system security?

- Operating systems provide various security features such as password-protected system accounts, a **firewall**, virus scanning and **file encryption**
- Password-protected system accounts are a very common feature in operating systems

- System accounts can also be restricted from performing certain actions, e.g. editing network settings, installing unapproved software, changing the account settings of other users



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User Management

What is user management?

- User management is a process carried out by the operating system enabling different users to log onto a computer
- The OS is able to maintain settings for individual users, such as desktop backgrounds, icons and colour schemes
- A system administrator is able to allocate different **access rights** for different users on a network



Worked Example

Ella uses her computer to create artwork for a magazine

Ella makes use of system software.

One type of system software is the operating system.

Identify and describe **two functions** of an operating system [6]

How to answer this question

- Break down the 6 marks, 1 mark each for identifying a function of the operating system. For each function you need to make 2 points about how they work

Answer

- Memory management
 - Allocates memory to programs currently in use
 - Gets data from RAM
 - Stores data in RAM
- File management
 - Creating/editing/renaming files
 - Creating/editing/renaming folders
 - Movement of files/folders



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Hardware, Firmware & the Operating System

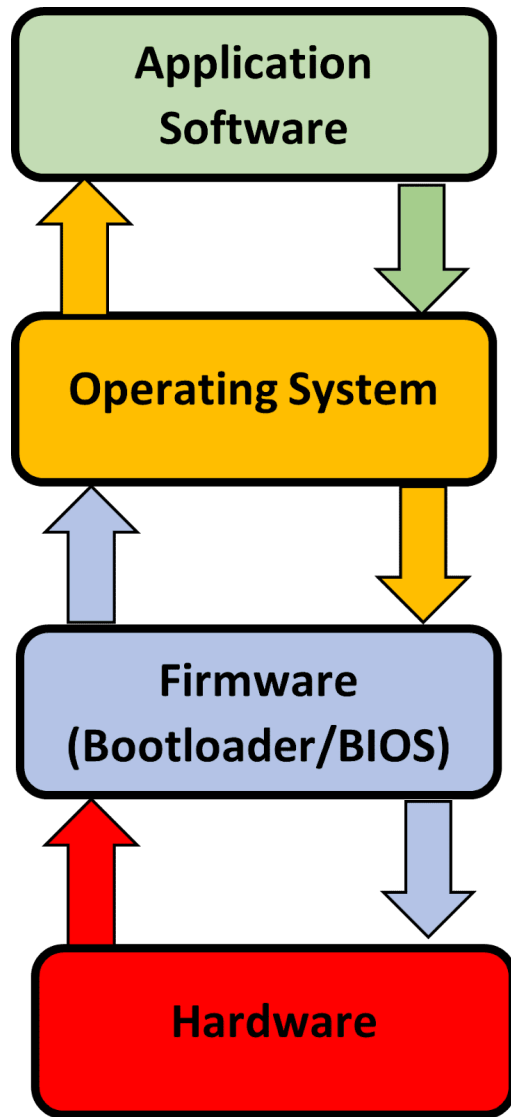
Hardware, Firmware & the Operating System

How does application software, the operating system and hardware communicate?

- **Application software** talks to the **operating system**, this allows it to **interact** with the **hardware**
- The hardware **processes** and **sends** the information to the **operating system** which **talks directly** to the applications software
- This process is **repeated** while application software is **in use**



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What is firmware?

- Firmware is **embedded directly in to the hardware of a device**, to make them function
- When a computer is turned on, it has to explore the **ROM** for its initial boot-up instructions, these are contained in a **Bootstrap loader**
- The initial process is handled by the **basic input/output system (BIOS)** which is known as **firmware**
- Once start-up is complete, instructions are sent to **RAM** to be processed by the operating system

- This layer ensures that hardware devices e.g. keyboard and mouse **are available** and can be **communicated directly by the operating system**
- Firmware **translates** between the **hardware** and the **software**



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Interrupts

Interrupts

What is an interrupt?

- An interrupt is a **signal for the CPU to stop** what it is **currently** doing and **do something else** as a higher priority
- The CPU is in a continuous loop of carrying out the fetch-decode-execute cycle, however there are occasions when this **needs to be interrupted**

How is an interrupt generated?

- An interrupt can be generated by hardware and software:
 - **Hardware** – this is caused by a hardware device such as a hardware failure
 - **Software** – this occurs when an application stops or requests services from the OS
- Interrupts are added to an area called the **interrupt service routine**
- The interrupt service routine **holds instructions** that will need to be fetched, decoded and executed to complete the commands of the **interrupt**
- The contents of the registers within the CPU cannot be lost by an interrupt, so contents are **copied to a reserved area in RAM** called a **stack**
- Contents are added to the **top of the stack**, which will **save them for later** retrieval when the interrupt is complete
- The interrupt will be **executed** instead of the original instructions

What are examples of hardware interrupts?

- **Hardware**
 - power button may have been pressed
 - moving the mouse
 - clicking an icon to open a new program
 - keyboard presses e.g. ctrl, alt, delete

What are examples of software interrupts?

- **Software**

- a program is not responding
- division by zero
- two processes trying to access the same memory location



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Examiner Tips and Tricks

The main focus of interrupts is the importance of the **interrupt service routine** and the role it plays in **moving current instructions to the stack** so that they can be interrupted **BUT** continue again afterwards



Worked Example

Describe the purpose of an interrupt in a computer system

[4]

Answer

Four from:

- Used to attend to certain tasks/issues
- Used to make sure that vital tasks are dealt with immediately
- The interrupt/signal tells the CPU/processor (that its attention is required)
- A signal that can be sent from a device (attached to the computer)
- A signal that can be sent from software (installed on the computer)
- The interrupt will cause the OS/current process to pause
- The OS/CPU/ISR will service/handle the interrupt
- They have different levels of priority
- After the interrupt is serviced, the (previous) process is continued
- It enables multi-tasking to be carried out on a computer
- A valid example of an interrupt e.g. 'out of paper' message for a printer